THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The largest number to be subtracted from 5091 to make it a perfect square is: 1

(a) 51 **(b)** 50 **(c)** 52 **(d)** 53

2. The difference of the squares of two consecutive natural numbers is 89. The numbers are: 1

(a) 44 and 46 **(b)** 45 and 46 **(c)** 44 and 45 **(d)** 43 and 44

3. The number of digits in the square root of a 6-digit perfect square is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 2

4. $\sqrt[3]{-1331}$ is: 1

(a) -11 **(b)** -12 **(c)** 12 **(d)** 11

5. The ones digit of 238^3 is: 1

(a) 2 **(b)** 4 **(c)** 6 **(d)** 8

6. The value of $\sqrt{529} + \sqrt{64}$ is: 1

(a) 31 **(b)** 184 **(c)** 15 **(d)** 32

7. The number of perfect squares between 500 and 2000 (both excluded) is: 1

(a) 24 **(b)** 23 **(c)** 22 **(d)** 21

8. The smallest number to be added to 3456 to make it a perfect square is: 1

(a) 26 **(b)** 27 **(c)** 24 **(d)** 25

9. **Assertion (A):** The cube of a negative integer is negative. 1

Reason (R): The cube of every integer is positive.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The number 392 is a perfect cube. 1

Reason (R): A number is a perfect cube if every prime in its prime factorisation has an exponent that is a multiple of 3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The difference between the squares of two consecutive numbers is 105. Find the numbers. 2

12. Find the number of non-square numbers between 8^2 and 11^2. 2

13. Find the smallest number to be added to 8140 to make it a perfect square. What is the square root of the result? 2

14. The difference between the squares of two consecutive numbers is 137. Find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A square garden has area 2704 m². Fencing costs ₹30 per metre. Find the total cost of fencing it. 3

16. A solid cube of side 9 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

17. A rectangular hall 45 m by 60 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

OR

A square garden has area 4096 m². Fencing costs ₹25 per metre. Find the total cost of fencing it. 3

18. A solid cube of side 10 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

OR

A rectangular hall 54 m by 72 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mohan, a shopkeeper in Indore, packs sweets in a cubical box of volume 2744 cm^3. He also has 1350 cm^3 of clay that he wants to mould into a cube. 5

(i) Find the edge of the box.

(ii) Find the total surface area of the box.

(iii) How many cubes of edge 2 cm fit in the box?

(iv) Find the smallest number by which 1350 must be multiplied so that the product is a perfect cube, and the edge of that cube.

20. Find the smallest square number that is divisible by each of 8, 14 and 21. 5

OR

Find the smallest cube number that is divisible by each of 9, 14 and 21. 5

END OF QUESTION PAPER — Total: 40 Marks